

Exercise set 1

Problem 1:

Consider the experiment of flipping an unbiased coin 5 independent times. Let H denote the outcome “head” and T the outcome “tail”.

- a) Compute the probability of HHTHT and THHHT exactly.
- b)^R Modify the R-code from the number of heads example in the lectures to simulate the probability of three heads occurring in the 5 trials.
- c) Can you also calculate this probability exactly? (Hint: Binomial distribution.)
- d)^R Redo point b) for a biased coin for which the probability of heads is 0.6. (Hint: See the help file for the `sample` function in R and set the `prob`-option.)
- e) Redo point c) for a biased coin for which the probability of heads is 0.6.

Problem 2:

For each of the following, determine the constant c such that $f(x)$ satisfies the condition of being the probability mass function (pmf) for a discrete random variable X .

- a) $f(x) = x/c$, $x = 1, 2, 3, 4$.
- b) $f(x) = c(x + 1)^2$, $x = 0, 1, 2, 3$.
- c)^R Depict each $f(x)$ from a) and b) as a bar graph.

Problem 3:

For the probability distributions in problem 2 find $E(X)$ and $E(g(X))$ with $g(x) = x^3$.

Problem 4:

A continuous random variable X has probability density function (pdf)

$$f(x) = \begin{cases} 4x(1 - x^2), & \text{for } 0 \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

- a) Calculate $P(X < 0.5)$, $E(X)$, $\text{Var}(X)$ and $\text{SD}(X)$.
- b) Find the cumulative distribution function (cdf), $F(x)$, and use this to calculate $P(X < 0.3)$ and $P(X > 0.7)$.
- c)^R Make a sketch of $f(x)$ and $F(x)$ in R.

Problem 5:

Let X be a continuous random variable with pdf

$$f(x) = \begin{cases} k(1 - x^2) & \text{for } -1 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

- a) Determine the constant k .
- b)^R Make a sketch of $f(x)$ in R.
- c) Calculate $P(X \leq 0.5)$.
- d) Let $Y = 1 + x^2$. Calculate $E(Y)$.

Problem 6:

The gas price on different days at two gas stations in Sandnes has been registered. Let X be the gas price at one of the stations and let Y be the gas price on the other station on a random day. Registrations of X and Y on 10 randomly chosen days are shown in the table below.

day i	1	2	3	4	5	6	7	8	9	10
x_i	23.89	23.39	22.20	24.35	24.10	23.39	23.96	24.15	23.69	22.57
y_i	23.99	23.39	22.65	24.25	23.99	23.09	23.66	24.25	23.36	22.57

- a)^R Plot the gas prices at the two stations against each other on a scatter plot using R.
- b)^R Calculate the empirical correlation between the gas prices and comment the result. (Hint: Use the `cor`-function in R.)

Problem 7:

In a blind test one wants to find out whether people can distinguish between the taste of Pepsi and Coca Cola. 10 persons are each given 5 samples to test drink. One of these samples contains Pepsi, the other four Coca Cola. The test people are not told which are which and they are asked to try to pick out the Pepsi sample. Let X be the number of correct identifications.

- a) What kind of a distribution does X have, if they in fact taste the same? Find $P(X \geq 3)$, $E(X)$ and $\text{Var}(X)$.
- b) Let us now imagine that the test goes on and on (with more than 10 persons if necessary) and let N be the number of persons needed for the first correct identification. What kind of distribution does N have and what is $E(N)$?

Problem 8:

The height of Norwegian men in their twenties follows a normal distribution with mean 180 cm and standard deviation 6.5 cm.

Use the R function `pnorm` to calculate the following probabilities:

- a)^R The probability that a man has a height of less than 175.
- b)^R The probability that a man has a height of more than 190.
- c)^R The probability that a man has a height between 170 to 180.

Use the R function `qnorm` to calculate:

- d)^R The height which is such that only 1% of the men have a higher height.

Problem 9:

Three investments give returns X , Y and Z characterized by $E(X) = E(Y) = E(Z) = 0.4$, $\text{Var}(X) = \text{Var}(Y) = \text{Var}(Z) = 1.0$. Moreover, there are the following correlations between these returns: $\rho(X, Y) = 0$, $\rho(X, Z) = 0.5$ and $\rho(Y, Z) = -0.5$:

- a) Consider the returns from the following four investments: (i) $2X$, (ii) $X + Y$, (iii) $X + Z$ and (iv) $Y + Z$. Compute the expectation and variance for each of these cases.

Which investment would you prefer if variance is taken as a measure of risk?